FROM RFM TO CLV

Buy-Till-You-Die and Gamma-Gamma models

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LEARNING OBJECTIVES

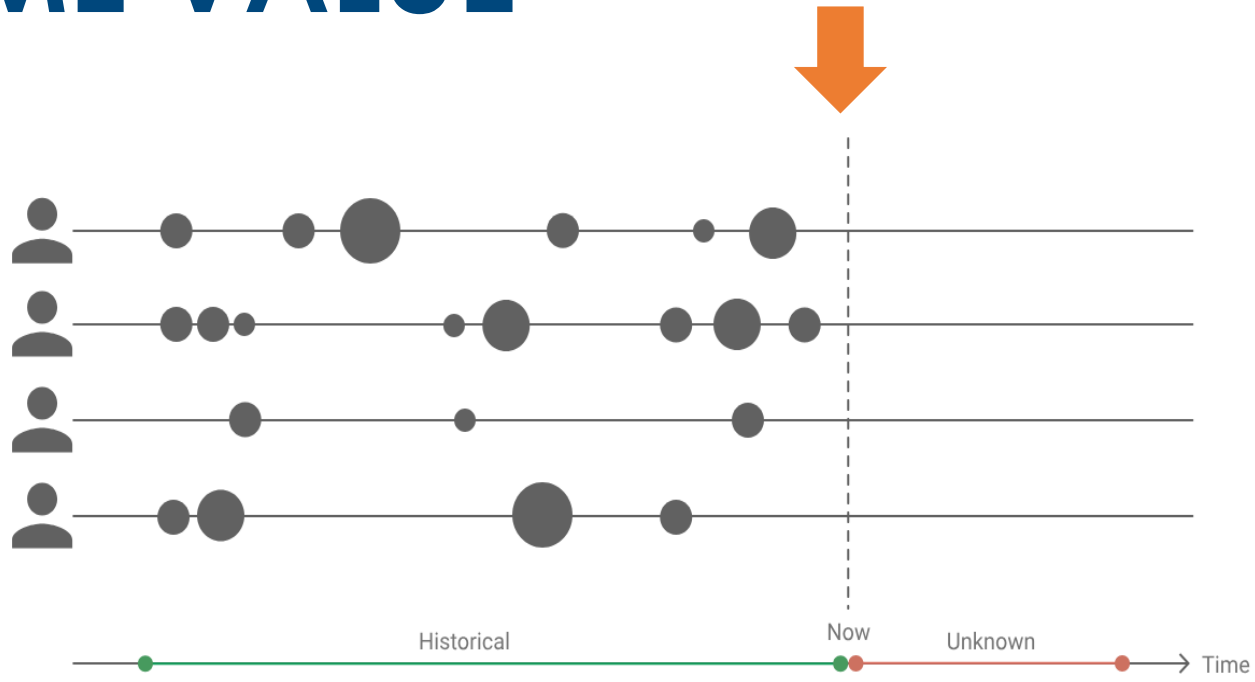
- Understand the structure of the CLV formula and its components
- Explain the behavioral assumptions behind Buy-Till-You-Die models
- Estimate a Buy-Till-You-Die model in R and interpret its outputs
- Explain why monetary value requires a separate model
- Estimate and interpret a Gamma–Gamma model in R
- Combine purchase and value models to obtain CLV estimates

CUSTOMER LIFETIME VALUE

CLV is the present value of all future profits a customer will generate for a business.

Hence, CLV is fundamentally about predicting forward *from partial history.*

The next formulas formalize this idea, but they do not solve it.



Past Present Future

FORMULA

This formula tells us *what we would do* **if** we knew the future

Every term in this formula refers to the future - and the future is *unknown*

The formula defines the target, *not the method*.

$$CLV = \sum_{t=0}^{T} \frac{(p_t - c_t) r_t}{(1+i)^t} - AC,$$

- $p_t - c_t$: profit *if* the customer buys at time t
- r_t: whether the customer is still active at time t
- $(1+i)^t$: discounting future money
- AC: what we already spent to acquire them

Ref.: Tudoran, A.A., Thomsen , C., & Thomasen , S. (2024). Understanding consumer behavior during and after a pandemic: Implications for customer lifetime value prediction models. Journal of Business Research, 174,.

WHY EXPECTATIONS APPEAR IN THE 2ND FORMULA

This is just a more *honest version* of the first formula. Here we admit that future value is uncertain

Because we don't know the future, we replace future value with ***expected future value***

So, we observe partial history, and from that we form *expectations about what will happen next*

$$LTV = \sum_{t=1}^{\infty} \frac{E[\widetilde{V}_t]}{(1+\delta)^{t-1}}$$

- $E[\tilde{V}_t]$ = expected future value in period t
- The tilde reminds us this is a **random variable**
- The summation runs to infinity because we do not know when the customer will churn

Ref.: Blattberg, C. B., Byung, D. K., & Scott, A. N. (2008). Database Marketing: Analyzing and Managing Customer

BUY-TILL-YOU-DIE (BTYD) & GAMMA–GAMMA MODELS

We break the problem ***into two parts***:

1. Will the customer buy again, and when? → BTYD models
2. If they buy, how much will they spend? → Gamma–Gamma model

CLV is *not* one model; it is a combination of models

BTYD × Gamma–Gamma = CLV

FROM FORMULA TO MODEL

CLV formulas define ***what*** to compute

The models define ***how*** to predict

History data → expectations/predictions → discounted value → CLV

BTYD MODELS

How many transactions will a customer make in a future period? (timing)

How long will the customer *remain active*? (survival)

Assumptions:

- Customers make purchases while "alive"
- Each customer has a latent purchase rate
- Customers may become inactive at some point
- We never observe churn directly — we infer it from purchase patterns

TWO MODELS

Model	Purchase process	Dropout process	Meaning
Pareto/NBD	Repeated purchases	*Continuous*-time dropout	Customer can churn at any moment
BG/NBD	Repeated purchases	*Discrete* dropout after purchases	Customer churns after a transaction

The difference between the two models is how we model customer inactivity.

STATISTICAL IMPLEMENTATION

Purchase rate heterogeneity as (**λ**) modelled→ Negative Binomial

Dropout timing (**μ**) modeled differently:

- Pareto/NBD: continuous time (models the time between events)
- BG/NBD: discrete after purchases (models the number of trials until success – "death")

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Understanding consumer behavior during and after a Pandemic: Implications for customer lifetime value prediction models

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ABSTRACT

Our study uses a cohort analysis to investigate Customer Lifetime Value (CLV) for customer cohorts acquired before and during the COVID-19 pandemic. The research estimates CLV in a continuous-time setting of customer transactions within the online grocery sector. Stochastic models are combined with the Gamma-Gamma spending model to predict CLV at individual and aggregate levels. The findings reveal the satisfactory fit of the models at both individual and aggregate levels. Combined with the Gamma-Gamma model, the MBG/NBD model stands out as the top performer, accurately classifying over 60 % of the best-CLV customers (top 10 % and 20 %). Cohort-based analyses outperform overall sample models in terms of out-of-sample errors. Furthermore, CLV prediction models differ between the customer cohorts analyzed. The models for the pre-COVID-19 cohort underestimate the cumulative CLV, whereas models for the COVID-19 cohort overestimate it. These discrepancies can relate to the shifting behavior of the COVID-19 and pre-COVID-19 customer cohorts.

1. Introduction

Customer lifetime value (CLV) is a fundamental concept in business operations, providing critical insights into long-term strategies and decision-making (Kumar & Reinartz, 2016; Kumar & Reinartz, 2018; Mosaddegh, Albadvi, Sepehri, & Teimourpour, 2021). It involves estimating the net present value of future profits from each customer and in total over time (Gupta et al., 2006; Kumar & Reinartz, 2016). To make these estimations, CLV models traditionally rely on transactional data collected from customers' credit cards, loyalty programs, or online accounts, assuming these data reflect consistent purchasing patterns. However, the business landscape can be subject to sudden and unforeseen disruptions, as exemplified by the COVID-19 pandemic, with possible implications for CLV models.

The COVID-19 crisis brought about significant alterations in consumer and business behavior, most notably a surge in e-commerce spending, followed by a gradual return to pre-pandemic norms in online and brick-and-mortar shopping categories by the summer of 2022 (Oblander & McCarthy, 2022). The effects of the COVID-19 pandemic's transformative impact continue to captivate researchers' attention (Hardt, Gluckstad, Hardt, & Glückstad, 2024; Mele et al., 2023; Meyer, Prescott, & Sheng, 2022; Pineiro-Chousa et al., 2023; Raj, Mukherjee, Jabbour, & Srivastava, 2022). Within this context, several researchers investigate whether the changes in consumer behavior after a major event represent enduring shifts or merely transient adjustments (Oblander & McCarthy, 2022; Sheth, 2020). Some research indicates that COVID-19 has had profound and enduring effects on customers' purchasing behavior, possibly making online shopping the new standard (Dwivedi et al., 2020; Roggeveen & Sethuraman, 2020). Other studies have unveiled notable changes in product preferences (Sim, Cho, Hwang, & Telang, 2022), purchasing motivation (Vázquez-Martínez, Morales-Mediano, & Leal-Rodríguez, 2021), and spending priorities (Kirk & Rifkin, 2020; Sheth, 2020; Vázquez-Martínez et al., 2021).

Considering this transformative event, this research aims to evaluate the performance of CLV models in the context of customer acquisition during the pandemic and the volatile behavioral patterns resulting from the pandemic. Such behavioral shifts can profoundly impact the performance of current CLV prediction models. Consequently, it becomes imperative to estimate how these dynamics, observed in pandemic-acquired customer groups, contrast with conventional e-commerce customers and how they influence CLV predictions. The main objective of this study is to analyze the impact of behavior changes on the

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Table 1
A chronological review of relevant literature.

Authors	Objective	Modeling	Industry
Venkatesan & Kumar (2004)	To evaluate the usefulness of CLV as a metric for customer segmentation and resource allocation	RM MC	High technology
Fader et al., (2005a)	To develop BG/NBD model, an alternative to Pareto/NBD model	Pareto/ NBDBG/NBD	CDs
Fader et al., (2005b)	To develop a new model that links RFM metric with CLV using Iso-value curves	Pareto/NBD GG	CDs
Kumar, Shah, & Venkatesan (2006)	To develop a CLV metric useful for maximizing profitability in the apparel industry	GGG RM	Apparel, shoes, accessories
Fader, Hardie, & Jerath (2007)	To demonstrate the use of Pareto/NBD models for computing CLV	Pareto/NBD	Hypothetical case study
Batislam et al., (2007)	To develop MBG/NBD model and to compare predictive performance of probability models	Pareto/NBD BG/NBD MBG/NBD	Offline grocery retail
Wubben and Wangenheim, (2008)	To compare the quality of stochastic models when applied to managerial decision	Pareto/NBD BG/NBD	Airline, apparel, CDs
Abe (2009)	To extend Pareto/NBD model of customer-base analysis using a Hierarchical Bayesian model	Pareto/NBD Hierarchical Bayesian	CDs, department store (varied merchandize)
Glady et al. (2009)	To develop a modified Pareto/NBD that relaxes the independence assumption between the number of transactions and the average profit per transaction	Pareto/NBD Modified Pareto/NBD	Retail banking sector, CDs
Bemmaor & Glady (2012)	To develop a new CLV model using the gamma/Gompertz distribution (G/G)	Pareto/NBD G/G	CDs, US charity, gift catalog, non-profit organization, specialty catalog
Ma, Tan & Shu (2015)	To develop a continuous-time dynamic model	Pareto/NBD Markov chain	Video equipment retailer
Platzer and Reutterer (2016)	To develop a new generalization of Pareto/NBD: Pareto/gamma-gamma-gamma (GGG) model	Pareto/NBD Pareto/GGG	CDs, apparel and accessories, donations, groceries, dietary supplements, office supply
Jasek et al. (2018)	To empirically validate the predictive performance of stochastic models, extending Pareto/NBD with Gamma-gamma (GG) model	Pareto/NBD GG MC	Games, sports equipment, health products, erotic shops, health and beauty products, baby care, home decoration, interior design, fashion
Jasek et al. (2019)	To empirically validate the predictive performance of stochastic customer base models	Pareto/NBD Pareto/GGG BG/NBD BG/CNBD-k MBG/ NBDMBG/ CNBD-k	Same as Jasek et al. (2018)

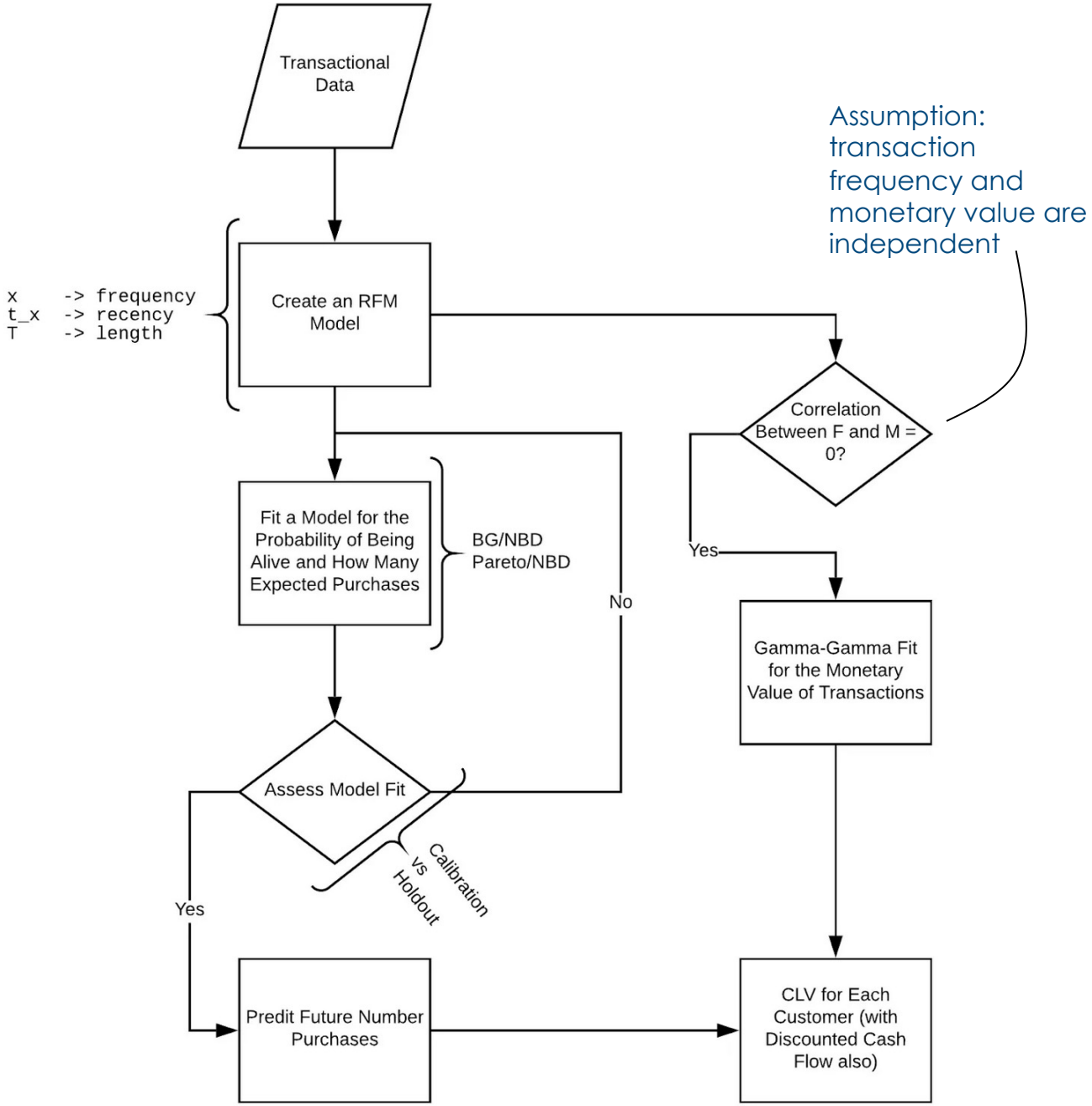
Table 1 (*continued*)

Authors	Objective	Modeling	Industry
Bachmann et al. (2021)	To extend Pareto/NBD model by modeling time-invariant and time-varying contextual factors	Pareto/NBD	Multichannel catalog merchant, electronics, sports goods
Reutterer et al. (2021)	To extend the BG/NBD by using the Erlang-k distribution for interpurchase times	BG/NBD	Same data as Platzer and Reutterer (2016)
Kim et al. (2022)	To empirically validate the performance of stochastic customer base models	Pareto/NBD Pareto/GGG BG/CNBD-k MBG/CNBD-k	Securities industry
Valendin et al. (2022)	To test deep learning as an alternative to BTYD	BTYD RNN	Electronics, multichannel merchant, CDNOW, groceries, yogurt, and sunscreen purchases
This study	To assess the impact of customer acquisition timing on CLV prediction accuracy for cohorts acquired before and during the COVID-19 pandemic	Pareto/NBD BG/ NBDMGB/ NBD GG	Online grocery retail

Note: BG/CNBD-k = Beta Geometric/Conditional Negative Binomial Distribution model with a parameter k; BG/NBD = Beta Geometric/Negative Binomial Distribution model; BTYD ="Buy 'Til You Die" models; GG = Gamma-gamma spending model; GGG = Generalized Gamma Distribution model; G/G = Gamma/Gompertz distribution model; MBG/NBD = Modified Beta Geometric/Negative Binomial Distribution model; MC = Markov Chain model; Pareto/NBD = Pareto/Negative Binomial Distribution model; Pareto/GGG = Pareto/Generalized Gamma Distribution model; RM = regression model; RNN = Recurrent Neural Networks model.

PROCESS FLOW

- This slide shows the full CLV pipeline
- We start with raw transactions, summarize behavior, model future purchasing, model spending, and combine everything into a discounted value



Source: https://ishla.medium.com/predicting-customer-lifetime-value-clv-using-probabilistic-model-0330efe4e4b3

DATA

- We start with **transactional data**, but we deliberately throw most of it away
- The remaining variables are perfectly reasonable for other analyses (e.g. service quality, customer profiling etc.)
- BTYD models only care about three things:

  who bought
  when they bought
  how much they spent

SELECTED VARIABLES

- Customer identifier (user_id)
- Transaction date (date)
- Transaction value (sales)

```
>   variable.names(completedataclean)
 [1] "user_id"                        "id"
 [3] "active_at"                      "address_postalCode"
 [5] "amount_of_bags"                 "boosted"
 [7] "completed"                      "completed_at"
 [9] "date"                           "date_from_time"
[11] "date_to_time"                   "days_since_last_job"
[13] "delivered_at"                   "delivery_guy_age"
[15] "delivery_guy_gender"            "delivery_guy_id"
[17] "delivery_guy_jobs"              "delivery_guy_rating"
[19] "delivery_guy_payed_amount"      "delivery_price"
[21] "expired"                        "expired_at"
[23] "items_in_basket"                "items_sold_out"
[25] "min_between_taken_and_delivered" "nearest_meta_store_id"
[27] "sales"                          "started_at"
[29] "taken_at"                       "total_price"
[31] "user_age"                       "user_gender"
[33] "user_jobs"                      "user_rating"
[35] "using_web"                      "region_id"
[37] "region_name"                    "completefraction"
[39] "deliveredintimeminutes"         "deliveredintime"
>
```

DATA PRE-PROCESSING

Merge transactions on the *same date*

Data cleaning required, incl. *remove outliers*

Re-name the main variables as

"cust"

"date"

"sales"

as the model (library "BTYD", in R) expects these names

cust	date	sales
894	2017-12-03	254.00
894	2017-12-02	254.00
1095	2017-02-01	254.00
1095	2017-07-21	254.00
1095	2017-05-28	254.00
1816	2020-04-04	255.67
1816	2021-03-22	481.30
1816	2020-12-27	300.53
1816	2021-02-01	453.05
1836	2019-06-21	95.40
4802	2019-07-16	101.70
4802	2019-01-02	152.95
4802	2018-01-02	252.00
4802	2018-12-13	89.90
4802	2018-09-14	47.25
5163	2017-09-10	254.00
6645	2021-07-07	300.25
6645	2018-08-13	87.35

CALIBRATION MATRIX

A calibration matrix is a structured *summary of each customer's past purchasing behavior*, expressed in a way the probabilistic model can use

The calibration matrix moves us ***to the customer level***. CLV is defined at the customer level, not the transaction level

```
data2                    List of  3
  $ cal      :List of 2
   ..$ cbs: num [1:669, 1:3] 1 2 2 0 4 0 7 66 1 1 ...
   .. ..- attr(*, "dimnames")=List of 2
   .. .. ..$ : chr [1:669] "894" "1095" "1816" "1836" ...
   .. .. ..$ : chr [1:3] "x" "t.x" "T.cal"
   ..$ cbt: num [1:669, 1:945] 0 0 0 0 0 0 0 0 0 0 ...
   .. ..- attr(*, "dimnames")=List of 2
   .. .. ..$ : chr [1:669] "894" "1095" "1816" "1836" ...
   .. .. ..$ : chr [1:945] "2017-01-21" "2017-01-27" "2017-02-02" "2017-02…
  $ holdout  :List of 2
   ..$ cbt: num [1:669, 1:388] 0 0 0 0 0 0 0 0 0 0 ...
   .. ..- attr(*, "dimnames")=List of 2
   .. .. ..$ : chr [1:669] "894" "1095" "1816" "1836" ...
   .. .. ..$ : chr [1:388] "2021-02-15" "2021-02-16" "2021-02-17" "2021-02…
   ..$ cbs: num [1:669, 1:2] 0 0 1 0 0 0 13 3 0 0 ...
   .. ..- attr(*, "dimnames")=List of 2
   .. .. ..$ : chr [1:669] "894" "1095" "1816" "1836" ...
   .. .. ..$ : chr [1:2] "x.star" "T.star"
  $ cust.data:'data.frame':      669 obs. of  5 variables:
   ..$ cust       : int [1:669] 894 1095 1816 1836 4802 5163 6645 6658 683…
   ..$ birth.per  : Date[1:669], format: "2017-12-02" "2017-02-01" "2020-0…
   ..$ first.sales: num [1:669] 254 254 255.7 95.4 252 ...
   ..$ last.date  : Date[1:669], format: "2017-12-03" "2017-07-21" "2021-0…
   ..$ last.sales : num [1:669] 254 254 453.1 95.4 101.7 ...
```

CALIBRATION MATRIX

- Notice that the data are split into two parts: ***calibration*** and ***holdout***
- The calibration period is used to estimate the *model parameters*. The holdout period is used to evaluate predictions
- Once we have this calibration matrix, we can estimate a model

```
data2                    List of  3
  $ cal      :List of 2
   ..$ cbs: num [1:669, 1:3] 1 2 2 0 4 0 7 66 1 1 ...
   .. ..- attr(*, "dimnames")=List of 2
   .. .. ..$ : chr [1:669] "894" "1095" "1816" "1836" ...
   .. .. ..$ : chr [1:3] "x" "t.x" "T.cal"
   ..$ cbt: num [1:669, 1:945] 0 0 0 0 0 0 0 0 0 0 ...
   .. ..- attr(*, "dimnames")=List of 2
   .. .. ..$ : chr [1:669] "894" "1095" "1816" "1836" ...
   .. .. ..$ : chr [1:945] "2017-01-21" "2017-01-27" "2017-02-02" "2017-02…
  $ holdout  :List of 2
   ..$ cbt: num [1:669, 1:388] 0 0 0 0 0 0 0 0 0 0 ...
   .. ..- attr(*, "dimnames")=List of 2
   .. .. ..$ : chr [1:669] "894" "1095" "1816" "1836" ...
   .. .. ..$ : chr [1:388] "2021-02-15" "2021-02-16" "2021-02-17" "2021-02…
   ..$ cbs: num [1:669, 1:2] 0 0 1 0 0 0 13 3 0 0 ...
   .. ..- attr(*, "dimnames")=List of 2
   .. .. ..$ : chr [1:669] "894" "1095" "1816" "1836" ...
   .. .. ..$ : chr [1:2] "x.star" "T.star"
  $ cust.data:'data.frame':      669 obs. of  5 variables:
   ..$ cust       : int [1:669] 894 1095 1816 1836 4802 5163 6645 6658 683…
   ..$ birth.per  : Date[1:669], format: "2017-12-02" "2017-02-01" "2020-0…
   ..$ first.sales: num [1:669] 254 254 255.7 95.4 252 ...
   ..$ last.date  : Date[1:669], format: "2017-12-03" "2017-07-21" "2021-0…
   ..$ last.sales : num [1:669] 254 254 453.1 95.4 101.7 ...
```

RECALL

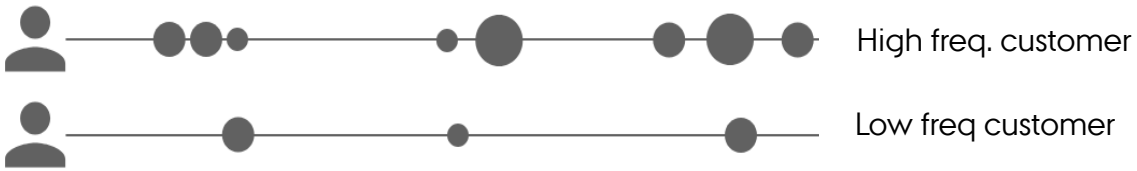
Model	Purchase process	Dropout process	Meaning
Pareto/NBD	Repeated purchases	*Continuous*-time dropout	Customer can churn at any moment
BG/NBD	Repeated purchases	*Discrete* dropout after purchases	Customer churns after a transaction

Now, how these assumptions translate into parameters and predictions???

FROM BEHAVIORAL ASSUMPTIONS TO MODEL PARAMETERS

Purchase process

Customers buy repeatedly while they are active
Different customers buy at different rates



NBD captures heterogeneity in purchase intensity across customers

λ (lambda): individual purchase rate
r, α: how much purchase rates differ across customers

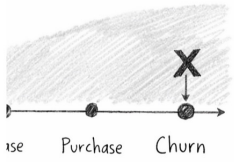
Dropout process

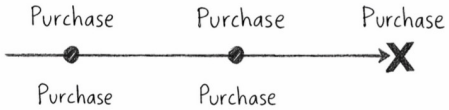
i. Pareto/NBD

Continuous-time Dropout (Pareto)

ii. BG/NBD

Discrete dropout (Beta-Geometric)





Pareto and BG captures heterogeneity in dropout timing across customers

s, β: how quickly dropout risk increases over time

α, b: how likely a customer is to drop out after a transaction

MEANING

Purchase process

r

→ Controls how concentrated or spread out purchase counts are
→ *Higher r: customers behave more similarly*

α

→ Scales the overall purchase intensity
→ *Higher α: lower average purchase rates*

MEANING

Dropout process

s - (shape of time-based churn risk)
→ Shapes *how sharply* dropout risk increases with time
→ *Higher s: more gradual increase in churn risk*

β – (how long customers tend to stay)
→ Sets the overall level of dropout risk
→ *Higher β: customers stay active longer on average*

a – (how easy it is to lose)
→ captures heterogeneity in churn propensity
→ *Higher a: higher probability of churning immediately after a transaction*

b – (how easy it is to keep)
→ Sets how likely customers tend to remain active
→ *Higher b: customers are more likely to continue after a transaction*

R OUTPUT

```
>   params.pareto
[1] 0.5172911 2.5238145 0.3383925 2.1250836
```

r α s β

```
> params.bg  # Print the estimated parameters
       p1        p2        p3        p4
0.2209466 0.9520612 0.3990803 1.5095206
```

r α a b

REPORTING RESULTS

Source: Tudoran et al. (2024)

- Whether a parameter value is considered "high" or "low" will depend on the context
- Analysts typically compare these parameters across different segments within the dataset or against historical data or industry to draw meaningful conclusions about customer behavior changes or differences.

Table 5
Estimated model parameters in the overall sample and by cohort.

Parameter	Pareto/NBD	BG/NBD
Overall sample		
r	0.49	0.23
α	2.10	0.91
s	0.25	
β	1.26	
a		0.27
b		1.14
k		
LL	−287,206	−287,476
$E(\lambda) = r/\alpha$	0.23	0.25
$E(\mu) = s/\beta$	0.20	
$E(p) = a/(a+b)$		0.23
Pre-COVID-19 cohort		
r	0.46	0.24
α	3.44	1.66
s	0.31	
β	5.57	
a		0.32
b		1.89
k		
LL	−157,390	−157,510
$E(\lambda) = r/\alpha$	0.13	0.14
$E(\mu) = s/\beta$	0.06	
$E(p) = a/(a+b)$		0.14
COVID-19 cohort		
r	0.68	0.22
α	2.34	0.74
s	0.35	
β	0.91	
a		0.36
b		1.16
k		
LL	−97,977	−98,511
$E(\lambda) = r/\alpha$	0.29	0.3
$E(\mu) = s/\beta$	0.38	
$E(p) = a/(a+b)$		0.24

MODEL PERFORMANCE

- Root means square error (RMSE)
- Normalized mean square error (NMSE)

Table 6
Stochastic models performance in the validation sample.

	RMSE	MAE	NMSE
Overall sample			
Pareto/NBD	10.114	3.712	0.505
BG/NBD	10.148	3.713	0.509
MBG/NBD	10.053	3.640	0.499
Pre-COVID-19 cohort			
Pareto/NBD	6.313	2.302	0.516
BG/NBD	6.328	2.330	0.517
MBG/NBD	6.309	2.303	0.515
COVID-19 cohort			
Pareto/NBD	8.067	3.312	0.704
BG/NBD	8.217	3.359	0.730
MBG/NBD	7.903	3.193	0.676

NMSE values below 1 indicate that the stochastic models outperform the baseline (model using the mean value of the target variable in the training sample)

GAMMA-GAMMA MODEL

It estimates the *expected average transaction value* of a customer

Notice, this is not total spending, just value per transaction

ASSUMPTIONS

- Each customer has a **stable spending level** over time
- How much a customer spends (M) is **independent** of how often they purchase (F)

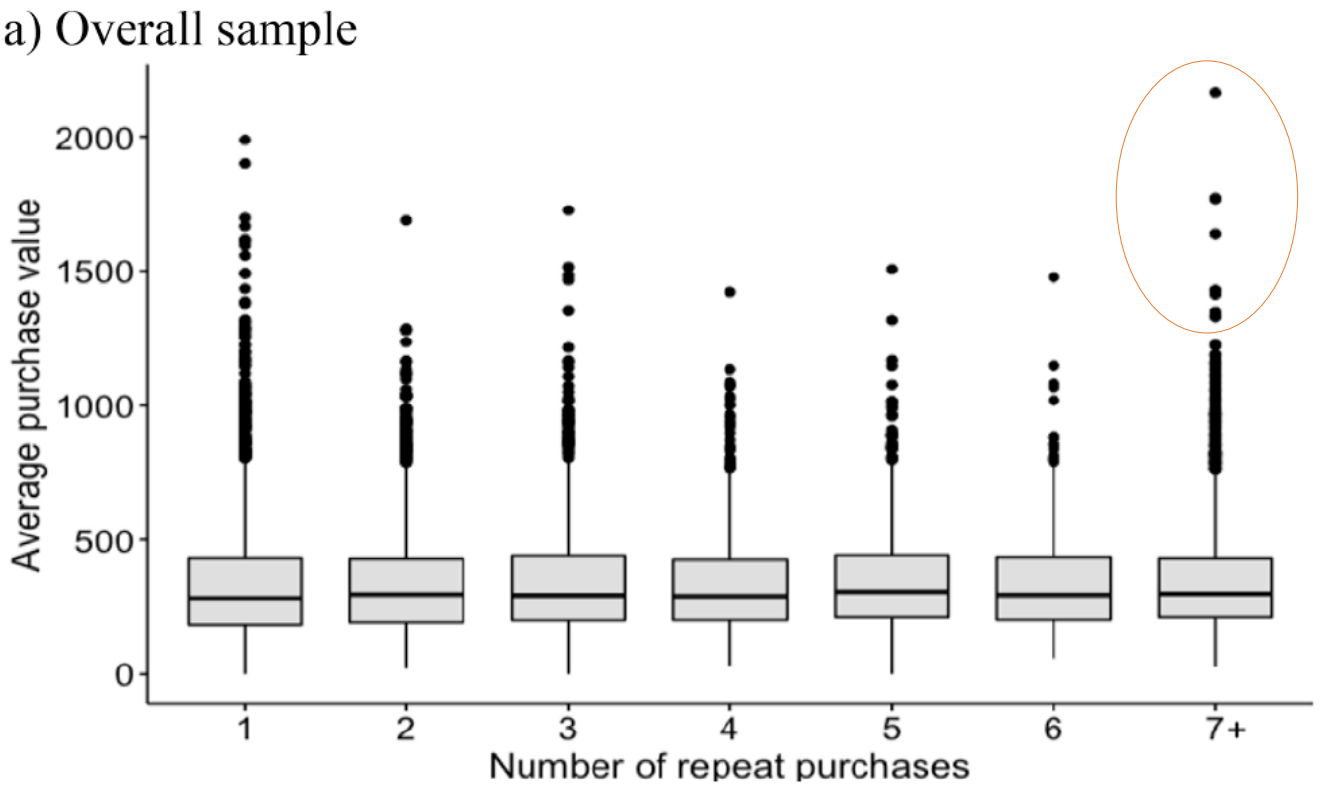
  → It allows us to treat purchase frequency (F) and transaction value (M) as separate processes.

INDEPENDENCY M & F

Before we use such a model, we should *check* whether this assumption is plausible in our data

For this data, what we see is that *the median* spending level is remarkably similar across frequency groups

Of course, there are high-spending *outliers* in every group. But the model requires no systematic relationship between M and F



Customers who buy once do not systematically spend more or less per transaction than customers who buy many times.

Heavy buyers buy more often — not more expensively.

Source: Tudoran et al. (2024)

MODEL PARAMETERS

The Gamma–Gamma model separates
variation *within* a customer's transactions
from
variation *between* customers' average spending levels

MODEL PARAMETERS

p — Shape of transaction-level spending

→ Controls how *uneven* individual purchases are for a customer

→ Higher *p*: transactions are more similar in size

→ Lower *p*: many small purchases, a few very large ones

q — Heterogeneity across customers

→ Controls how much customers differ in their *average* spending

→ Higher *q*: customers are more similar to each other

→ Lower *q*: big differences between low- and high-spending customers

γ — Overall spending scale

→ Sets the typical size of a transaction in the population

→ Higher γ: higher average transaction value overall

R OUTPUT

```
> params_gamma
[1]   3.497522   5.785227 471.267833
```

p q γ

REPORTING RESULTS

Table 7
Parameters for Gamma-Gamma spending model.

Model parameter	Overall sample	Pre-COVID-19 cohort	COVID-19 cohort
p	2.757	1.907	4.385
q	4.977	5.244	4.79
γ	500.373	725.817	316.328

Source: Tudoran et al. (2024)

CLV MODEL

FROM ESTIMATED COMPONENTS TO CLV

CLV is obtained by multiplying predicted future purchase counts with predicted transaction value and discounting the resulting cash flows to the present

$$\mathrm{CLV}_i = \sum_{t=1}^{T} \frac{\underbrace{\mathbb{E}[N_{i,t}]}_{\text{BTYD: how many purchases}} \times \underbrace{\mathbb{E}[M_i]}_{\text{Gamma–Gamma: value per purchase}}}{(1+d)^t}$$

BTYD and Gamma–Gamma predict behavior and value; discounting is applied afterward to translate predictions into present value. The discount rate is chosen by the analyst, not estimated from data

R OUTPUT & REPORTING

```
head(cal.cbs.gamma.test[, c(14:16)])

# CLV.pred.pareto CLV.pred.bg CLV.real
# 1        0.000000    0.000000      0.0
# 2        2.907413    8.722238      0.0
# 3      883.167227  692.946593    481.3
# 4       10.333488   48.222945      0.0
# 5       93.703514   95.540838      0.0
# 6        3.444496   24.111473      0.0
```

Table 8
CLV models performance in the validati

	Individual level	
	RMSE	MAE
1. Benchmark models		
1.1. Mean	5,035	1,497
1.2 Status quo	5,028	1,498
2. CLV prediction models for overall sample		
2.1 Pareto/NBD + GG	3,490	1,252
2.2 BG/NBD + GG	3,497	1,250
2.3 MBG/NBD + GG	3,462	1,225
3. CLV prediction models for pre-COVID-19 c		
2.1 Pareto/NBD + GG	2,419	792
2.2 BG/NBD + GG	2,425	800
2.3 MBG/NBD + GG	2,418	1,002
4. CLV prediction models for COVID-19 coho		
2.1 Pareto/NBD + GG	3,124	1,164
2.2 BG/NBD + GG	3,153	1,179
2.3 MBG/NBD + GG	3,064	1,164

Source: Tudoran et al. (2024)

NEXT LECTURE

APPLICATION 2

DOWNLOAD APPLICATION AND DATA FROM BRIGHTSPACE

REFERENCE

- Tudoran, A. A., Thomsen , C., & Thomasen , S. (2024). Understanding consumer behavior during and after a pandemic: Implications for customer lifetime value prediction models. *Journal of Business Research*, *174*, Artikel 114527. https://doi.org/10.1016/j.jbusres.2024.114527

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